

Feature Based Dimension Reduction for Image Classification An ASL Gesture Case Study

Contents

1	Hand-Landmark Feature Representations and Invariance Properties	1
1.1	Image-Plane Coordinate Representation	1
1.2	Translated Landmark Coordinate Representation	2
1.3	Scaled Landmark Coordinate Representation	2
1.4	Three-Dimensional Coordinate Representation	3
1.5	Pairwise Euclidean Feature Representation	4
1.6	Angle-Based Feature Feature Representation	5
1.7	Hybrid Feature Representations	6
1.8	Summary of Invariance Properties	6

1 Hand-Landmark Feature Representations and Invariance Properties

The feature representations considered in this paper are motivated by the core idea that different transformations of the detected landmark output $L(I_i)$ embed different geometric invariances. The purpose of this section is not only to define these feature spaces but also to explain the invariance properties that motivate them

For a given image I_i let the detected two-dimensional landmarks be represented as points

$$p_j = (x_j, y_j) \in \mathbb{R}^2, \quad j = 0, 1, \dots, 20$$

Different feature maps ϕ will transform this set of points into different feature vectors. In image classification, whilst some variation in the data is typically meaningful most are incidental. For this case study the relative configuration of the fingers, joints and palms is meaningful. However, the absolute location of the hand within the frame, the size of the hand and small changes in the camera distance would infact be nuisance factors. A useful representation should preserve the geometric information needed for the classification whilst minimizing the sensitivity to irrelevant variations. The feature representations below are motivated by their ability to encode hand geometry whilst controlling for different typed of irrelevant variation.

1.1 Image-Plane Coordinate Representation

The simplest landmark representation uses the image-plane coordinated of each detected landmark. This representation derived in ?? is given by

$$\phi_{xy}(L(I_i)) = (x_0, y_0, x_1, y_1, \dots, x_{20}, y_{20})^\top \in \mathbb{R}^{42}$$

This preserves the direct spatial position of each landmark in the image and is the most widely used feature space. It is the the simple, low-dimensional and directly interpretable since each feature corresponds the horizontal and vertical position of a specific landmark. However, this representation is sensitive of the global position of the hand in the image. To illustrate this, let's suppose the entire hand is shifted by a fixed vector

$$a = (a_1, a_2) \in \mathbb{R}^2.$$

Then each landmark becomes

$$p'_j = p_j + a$$

In coordinates, this means

$$(x_j, y_j) \mapsto (x_j + a_1, y_j + a_2)$$

The transformed coordinate representation is

$$\phi_{xy}(L(I_i))' = (x_0 + a_1, y_0 + a_2, \dots, x_{20} + a_1, y_{20} + a_2)^\top$$

In general,

$$\phi_{xy}(L(I_i))' \neq \phi_{xy}(L(I_i)).$$

Therefore, raw image-plane coordinated are not translation invariant which means that a classifier trained on this feature space may partially learn where the hand appears in the frame rather than only learn the shape of the gesture.

1.2 Translated Landmark Coordinate Representation

To reduce sensitivity to the global image location of the hand, we define a translated coordinate representation. $\forall I_i \in \mathcal{D}$ and for each landmark j , we define the translated coordinates by subtracting the coordinate-wise minima among the landmarks in that image:

$$x_{ij}^{\text{trans}} = x_{ij} - \min_{0 \leq r \leq 20} x_{ir},$$

and

$$y_{ij}^{\text{trans}} = y_{ij} - \min_{0 \leq r \leq 20} y_{ir}$$

The translated coordinate representation is then

$$\phi_{\text{trans}}(L(I_i)) = (x_0^{\text{trans}}, y_0^{\text{trans}}, x_1^{\text{trans}}, y_1^{\text{trans}}, \dots, x_{20}^{\text{trans}}, y_{20}^{\text{trans}})^{\top} \in \mathbb{R}^{42}$$

This representation removes the global position of the hand, and we can now explicitly show this. Suppose each landmark is shifted by the same vector $a = (a_1, a_2)$. Then

$$x'_j = x_j + a_1$$

The minimum transformed x -coordinate is

$$\min_{0 \leq r \leq 20} x'_r = \min_{0 \leq r \leq 20} (x_r + a_1) = \left(\min_{0 \leq r \leq 20} x_r \right) + a_1, \quad \text{since } a_1 \in \mathbb{R} \text{ independent of } r$$

Therefore,

$$\begin{aligned} (x'_j)^{\text{trans}} &= x'_j - \min_{0 \leq r \leq 20} x'_r \\ &= (x_j + a_1) - \left(\min_{0 \leq r \leq 20} x_r + a_1 \right) \\ &= x_j - \min_{0 \leq r \leq 20} x_r \\ &= x_j^{\text{trans}} \end{aligned}$$

The same argument can clearly be generalized to the y -coordinates. This invariance to the global translation of the hand is incredible useful as the location within the gram is typically not employed when classifying a gesture. However, this representation is still sensitive to the apparent size of the hand. If the hand appears larger or smaller, the translated coordinate values also change accordingly.

1.3 Scaled Landmark Coordinate Representation

To reduce the sensitivity to hand size, we define a scaled representation. We first define the horizontal and vertical ranges of the detected landmarks

$$w = \max_{0 \leq r \leq 20} x_r - \min_{0 \leq r \leq 20} x_r \quad \text{and} \quad h = \max_{0 \leq r \leq 20} y_r - \min_{0 \leq r \leq 20} y_r$$

Assuming $w > 0$ and $h > 0$, define

$$x_j^{\text{scale}} = \frac{x_j - \min_{0 \leq r \leq 20} x_r}{w},$$

and

$$y_j^{\text{scale}} = \frac{y_j - \min_{0 \leq r \leq 20} y_r}{h}$$

The scaled coordinate representation is

$$\phi_{\text{scale}}(L(I_i)) = (x_0^{\text{scale}}, y_0^{\text{scale}}, x_1^{\text{scale}}, y_1^{\text{scale}}, \dots, x_{20}^{\text{scale}}, y_{20}^{\text{scale}})^\top \in \mathbb{R}^{42}$$

This first translated the hand to remove the global image position and then rescales the coordinated using the landmark bounding box. This feature space therefore reduces the sensitivity to both hand location and size. We now show the scale invariance property. Suppose every coordinate is multiplied by a positive constant $c > 0$

$$x'_j = cx_j, \quad y'_j = cy_j$$

Then

$$\min_r x'_r = \min_r (cx_r) = c \min_r x_r, \quad \text{and} \quad \max_r x'_r = \max_r (cx_r) = c \max_r x_r.$$

Therefore, the transformed width is

$$w' = \max_r x'_r - \min_r x'_r = c \max_r x_r - c \min_r x_r = c(\max_r x_r - \min_r x_r) = cw$$

Now compute the transformed scaled coordinate:

$$\begin{aligned} (x'_j)^{\text{scale}} &= \frac{x'_j - \min_r x'_r}{w'} \\ (x'_j)^{\text{scale}} &= \frac{cx_j - c \min_r x_r}{cw} \\ &= \frac{c(x_j - \min_r x_r)}{cw} \\ &= \frac{x_j - \min_r x_r}{w} \\ &= x_j^{\text{scale}} \end{aligned}$$

The same argument could be generalized to the y -coordinates. As shown in subsection 1.2 subtracting the coordinate minima is also translation invariant. Thus the scaled coordinated are invariant to translation and positive scaling.

1.4 Three-Dimensional Coordinate Representation

This feature space retains the relative depth coordinate in addition to the image-plane coordinates given by:

$$\phi_{xyz}(L(I_i)) = (x_0, y_0, z_0, x_1, y_1, z_1, \dots, x_{20}, y_{20}, z_{20})^\top \in \mathbb{R}^{63}$$

This is to test whether retaining relative depth improves the classification accuracy and robustness. Previous analysis have shown that some gestures have similiar image-plane landmark configurations but differ in how the fingers move towards or away form the camera. In such cases, it would be an implicit assumption that the z -coordinate would contain useful but it may also introduce additional noise. Therefore, comparing ϕ_{xy} and ϕ_{xyz} is one way to test whether the relative depth coordinate is beneficial for static ASL gesture recognition. However, we do need to be careful here as the z -coordinate must be interpreted carefully. In this setting, z is not the physical measurement of depth but the relative depth output produced by the landmark detector. Therefore, it should not be interpreted as distance in physical units. Instead, it provides additional relative geometric information about the hand.

1.5 Pairwise Euclidean Feature Representation

Coordinate-based representations describe where each landmark is located. However, another meaningful way to describe hand geometry would be through the distances between the landmarks. For two landmarks a and b , we define the pairwise Euclidean distance

$$d_{ab} = \|p_a - p_b\|_2 = \sqrt{(x_a - x_b)^2 + (y_a - y_b)^2}$$

Pairwise distances are translation invariant. To see this, suppose every landmark is shifted by a fixed vector $a_0 \in \mathbb{R}^2$. Then

$$p'_a = p_a + a_0, \quad p'_b = p_b + a_0$$

The transformed distance is

$$\begin{aligned} d'_{ab} &= \|p'_a - p'_b\|_2 \\ &= \|(p_a + a_0) - (p_b + a_0)\|_2 \\ &= \|p_a - p_b\|_2 = d_{ab} \end{aligned}$$

Thus, pairwise distances are invariant to global translation. However, pairwise distances are not automatically scale invariant. If every landmark is multiplied by $c > 0$, then

$$p'_j = cp_j$$

The transformed distance is

$$d'_{ab} = \|cp_a - cp_b\|_2 = \|c(p_a - p_b)\|_2$$

Using the homogeneity of the Euclidean norm,

$$d'_{ab} = c\|p_a - p_b\|_2 = cd_{ab}$$

Thus, distances scale by the same factor as the hand. To make distances scale invariant, one may normalize by a reference distance $d_{\text{ref}} > 0$:

$$\tilde{d}_{ab} = \frac{d_{ab}}{d_{\text{ref}}}$$

Under scaling,

$$d'_{ab} = cd_{ab}, \quad d'_{\text{ref}} = cd_{\text{ref}}$$

Therefore,

$$\tilde{d}'_{ab} = \frac{cd_{ab}}{cd_{\text{ref}}} = \frac{d_{ab}}{d_{\text{ref}}} = \tilde{d}_{ab}$$

Thus, normalized pairwise distances are both translation invariant and scale invariant. Distance features are interpretable because they describe the relative spacing between landmarks which in itself encodes the geometry of the hand. For example, distances between fingertips represent finger separation whilst distances from finger tips to the wrist can describe finger extension.

1.6 Angle-Based Feature Feature Representation

Angled-based features describe the bending or orientation of fingers using triples of landmarks. Let a , b and c be three landmark indices. The angle at landmark b is formed by the vectors

$$u = p_a - p_b, \quad v = p_c - p_b.$$

The angle is defined by

$$\theta_{abc} = \cos^{-1} \left(\frac{u \cdot v}{\|u\|_2 \|v\|_2} \right)$$

An angle-based representation is then

$$\phi_{\text{angle}}(L(I_i)) = (\theta_1, \theta_2, \dots, \theta_s)^\top \in \mathbb{R}^s,$$

where s is the number of landmark triples chosen. Angle features are translation invariant as they depend only on the difference between landmark positions. Suppose every point is shifted by $a_0 \in \mathbb{R}^2$. Then

$$p'_j = p_j + a_0.$$

The transformed vector u' is

$$u' = p'_a - p'_b = (p_a + a_0) - (p_b + a_0) = p_a - p_b = u$$

Similarly,

$$v' = p'_c - p'_b = (p_c + a_0) - (p_b + a_0) = p_c - p_b = v$$

Since $u' = u$ and $v' = v$, the angle is unchanged:

$$\theta'_{abc} = \theta_{abc}.$$

Angle features are also invariant to positive scaling. Suppose $p'_j = cp_j$, where $c > 0$. Then

$$u' = p'_a - p'_b = cp_a - cp_b = c(p_a - p_b) = cu,$$

and

$$v' = p'_c - p'_b = cp_c - cp_b = c(p_c - p_b) = cv$$

The transformed cosine ratio is

$$\frac{u' \cdot v'}{\|u'\|_2 \|v'\|_2} = \frac{(cu) \cdot (cv)}{\|cu\|_2 \|cv\|_2}$$

Using the dot product and norm properties,

$$(cu) \cdot (cv) = c^2(u \cdot v),$$

and

$$\|cu\|_2 \|cv\|_2 = (c\|u\|_2)(c\|v\|_2) = c^2\|u\|_2 \|v\|_2$$

Therefore,

$$\frac{u' \cdot v'}{\|u'\|_2 \|v'\|_2} = \frac{c^2(u \cdot v)}{c^2\|u\|_2 \|v\|_2} = \frac{u \cdot v}{\|u\|_2 \|v\|_2}$$

Thus,

$$\theta'_{abc} = \theta_{abc}$$

Angle-based features are therefore invariant to both translation and positive scaling. These features could prove to be rather useful since many gestures are distinguished by finger bending and relative orientation. However, angle features may lose some information about absolute finger spacing, which is why it might be important better put to use when integrated withing a hybrid representation.

1.7 Hybrid Feature Representations

A hybrid representation combines multiple feature types into a single feature vector. One possible hybrid representation is

$$\phi_{\text{hybrid}}(L(I_i)) = [\phi_{\text{scale}}(L(I_i)), \phi_{\text{dist}}(L(I_i)), \phi_{\text{angle}}(L(I_i))]$$

This representation combines normalized coordinate information, pairwise distance information, and angle-based information. The motivation for hybrid features is that different representations encode different aspects of hand geometry. Coordinate features preserve landmark positions, distance features encode relative spacing, and angle features describe finger bending. A hybrid representation may improve classification performance by combining complementary geometric information.

However, hybrid representations also increase the feature dimension. This may improve accuracy, but it may also increase redundancy or reduce interpretability. Therefore, hybrid features are useful for studying the tradeoff between predictive performance and representation complexity.

1.8 Summary of Invariance Properties

The feature representations above differ in the types of nuisance variation they remove. Table 1 summarizes these properties.

Representation	Translation Invariant	Scale Invariant	Primary Geometry
Image-plane coordinates	No	No	Absolute landmark position
Translated coordinates	Yes	No	Relative landmark position
Scaled coordinates	Yes	Yes	Normalized hand shape
3D coordinates	With z -shift	With z -scale	Image-plane position + relative depth
Pairwise distances	Yes	No	Relative landmark spacing
Normalized distances	Yes	Yes	Scale-free spacing
Angles	Yes	Yes	Finger bending/orientation
Hybrid features	Depends	Depends	Combined geometry

Table 1: Invariance properties of landmark-based feature representations.

This summary shows why feature construction is central to the paper. Choosing a feature representation ϕ determines which aspects of the image are preserved and which nuisance transformations are removed. Raw coordinates preserve direct spatial information, but they are sensitive to hand position and size. Translated and scaled coordinates remove some of this sensitivity. Distance and angle features encode relative hand geometry more directly. Hybrid features combine several geometric descriptions, but at the cost of increased dimension.

Thus, the goal is not simply to reduce the number of variables. The goal is to construct a lower-dimensional, geometrically meaningful representation that preserves class-relevant information while improving robustness and interpretability.